

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

November 2019

EC 471 Digital Image Processing

Date: 13.11.2019, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q 1 Answer the following **[07]**

(a) Define Image and Pixel. **[02]**

(b) Draw and explain components of a general purpose image processing system. **[05]**

Q - 2 (a) Explain RGB color model in brief. **[07]**

(b) Explain histogram equalization in detail. **[07]**

OR

(b) Explain Four important image transforms. **[07]**

Q – 3 Answer the following **[14]**

(a) What is color space? Mention its classification. **[05]**

(b) Compute Median Value of Marked Pixels using 3x3 Mask. **[04]**

$$\begin{bmatrix} 24 & 22 & 33 & 25 & 32 & 24 \\ 34 & 0 & 24 & 255 & 26 & 23 \\ 23 & 21 & 32 & 31 & 28 & 26 \end{bmatrix}$$

(c) What are the effects of Dilation process on image? **[05]**

OR

Q – 3 Answer the following **[14]**

(a) Explain Opening and Closing in details. **[05]**

- (b) The Input image A and structuring element B are shown below. Perform Erosion and Dilation on input image [04]

0	0	0	0	0	0	0
0	0	0	0	1	1	0
0	0	1	1	0	1	0
0	0	1	1	0	0	0
0	0	0	1	1	0	0
0	0	0	0	0	0	0

Image A

0	1	0
1	1	1
0	1	0

Structuring Element B

- (c) Which are the various applications of image Segmentation? Explain any one of them. [05]

SECTION – II

- Q - 4 The differentiation property of continuous Fourier transform is: [07]

$$\left(\frac{\partial}{\partial x}\right)^m \left(\frac{\partial}{\partial y}\right)^n f(x,y) \Leftrightarrow (j2\pi u)^m (j2\pi v)^n F(u,v)$$

- (a) Derive this for the first derivative in x direction (i.e. m=1, n=0).
 (b) Sketch by hand the magnitude of the equivalent filter H(u,v) such that $G(u,v) = H(u,v) F(u,v)$
 (c) The discrete derivative approximation is $f(x+1, y) - f(x,y)$. Find the equivalent filter H(u,v) in the frequency domain.
 (d) Plot (as a surface) the magnitude of filter H(u,v) using MATLAB.

- Q - 5 (a) Explain the Wiener filter and constrained least square filter. [07]
 (b) Write an algorithm for Adaptive median filter. [07]

OR

- Q - 5 (a)** Explain various noise models and compare their distribution function. [07]
- (b) Derive mask (size of 3×3) of second derivatives by Laplacian, for enhancement in spatial domain. Prepare the other modified mask of the same. [07]
- Q - 6 (a)** Explain the model of the image Degradation/Restoration process. [06]
- (b) Explain the second order derivative for edge detection and derive Laplacian edge operator [06]

OR

- Q – 6 (a)** Explain Image segmentation based on Thresholding. [06]
- (b) Compare Canny Edge detector with the Laplacian Edge detector. [06]
- Q – 6 (c)** Draw diagram for basic steps for filtering in frequency domain [02]